

NanoGT

A Computational Framework for Mapping Monogenic Rare Diseases
to Gene-Therapy Development Precedents

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40-disease proof-of-concept cohort · 14-dimension heuristic scoring (v2) · 39 diseases with ranked precedents · 1 packaging hard-fail (NF1)

Abstract

Background

Most monogenic rare diseases have no disease-modifying treatment, even when the causal gene is known. Gene therapy is partly a platform technology: vector tropism, manufacturing, route of administration, safety monitoring, and regulatory precedent from one programme can reduce uncertainty for another biologically adjacent disease. The unresolved problem is how to identify defensible precedents systematically rather than by ad hoc expert review.

Methods

NanoGT was implemented as a Python command-line framework that matches Orphanet-indexed monogenic diseases to a curated catalogue of approved or clinical-stage gene therapy programmes. Each disease-program pair is scored across fourteen dimensions: packaging fit, tissue tropism, protein class, pathway similarity, molecular mechanism/modality compatibility, inheritance compatibility, approval precedent, vector immunogenicity, therapeutic window, cross-correction potential, immune privilege, promoter availability, route-of-administration feasibility, and organelle targeting feasibility. Scores are summed to a raw maximum of 21 and normalised to a 10-point composite confidence score. The current dissertation cohort contains 40 diseases: two positive controls, two benchmark diseases with known gene-therapy precedents, one oversized-cargo stress test, one mitochondrial stress test, 24 pilot discovery diseases, six haploinsufficiency diseases, two repeat-expansion/silencing diseases, one genomic imprinting disease, and one two-hit haploinsufficiency disease.

Results

The framework generated scored precedents for 39 of 40 diseases; Neurofibromatosis type 1 received a packaging hard-fail because its 8,451 bp coding sequence exceeds the capacity of every vector in the catalogue. Among the 39 scored diseases, 32 received high-confidence top matches and 7 received medium-confidence top matches. Duchenne muscular dystrophy failed the ordinary full-length native-gene packaging gate but was ranked through an engineered micro-dystrophin precedent. Composite scores ranged from 5.9/10 to 9.9/10. Libmeldy/lentiviral haematopoietic stem-cell therapy dominated lysosomal and leukodystrophy-like diseases, BMN 307 supported hepatic metabolic disorders, retinal diseases clustered around Luxturna/CPCB-RPE1 precedents, and the non-LOF arm produced biologically coherent medium-confidence matches for haploinsufficiency, repeat-expansion, and imprinting diseases.

Conclusions

NanoGT currently supports dissertation-level proof-of-concept claims: it can reproduce obvious clinical precedents, identify coherent cross-disease precedent clusters, and expose cases where a standard single-vector gene-addition strategy is weak or physically infeasible. It should not be framed as a clinical recommendation engine. The next month of work must prioritise source verification, reference reading, limitation handling, figure quality, and report writing rather than further scope expansion.

Keywords: gene therapy, rare disease, monogenic disease, Orphanet, AAV, lentiviral vector, precedent matching, computational biology

Introduction

The Rare Disease Treatment Gap

Approximately 7,000 distinct rare diseases have been identified to date, many of which are monogenic — caused by pathogenic variants in a single gene [18,19]. Collectively, rare diseases affect hundreds of millions of people worldwide, yet only a minority have approved disease-modifying treatments [18,21]. The gap between the number of identifiable diseases and the number with effective therapies represents one of the most significant unmet needs in modern medicine.

The causes of this treatment gap are predominantly economic and logistical rather than biological. Rare disease drug development is expensive — regulatory costs, manufacturing scale-up, and clinical trial execution must be amortised across patient populations that may number only in the hundreds or low thousands globally. For the most extreme cases — diseases with a prevalence below 1 in 1,000,000, sometimes termed ultra-rare or nano-rare — the commercial return is vanishingly small relative to development costs, and these diseases are systematically deprioritised or abandoned by industry. Children and adults with these conditions have no approved treatment options, often no active clinical trials, and frequently no funded research programme at all.

Gene Therapy as a Potential Platform Solution

Among the therapeutic modalities that have emerged over the past two decades, gene therapy occupies a uniquely promising position for rare monogenic diseases. By delivering a functional copy of a defective gene directly to the affected cells, gene therapy addresses the root cause of loss-of-function disorders rather than managing downstream symptoms. The approval of Luxturna (voretigene neparvovec, 2017) for RPE65-associated retinal dystrophy, Zolgensma (onasemnogene abeparvovec, 2019) for spinal muscular atrophy, and Hemgenix (etranacogene dezaparvovec, 2022) for haemophilia B established proof of concept that single-administration gene therapy can produce durable correction across a range of tissue targets and disease types.

Critically, gene therapy is increasingly understood to function as a platform technology. The biological and regulatory work required to bring a gene therapy to market — establishing the safety of a vector serotype in a given tissue, validating a tissue-specific promoter, demonstrating an acceptable immunogenicity profile, qualifying the manufacturing process — does not disappear once a therapy is approved. This body of evidence becomes a shared resource: a subsequent gene therapy program using the same vector and delivery approach to treat a different disease in the same tissue can leverage this established precedent. FDA and EMA gene-therapy guidance documents recognise the importance of vector quality, non-clinical safety, clinical monitoring, and long-term follow-up evidence [34–36]. The implication is significant: for many untreated rare diseases, the foundational work needed to begin clinical development may already have been done, just for a different disease.

The Precedent Identification Problem

Realising this opportunity in practice requires answering a question that is deceptively difficult: for a given untreated rare disease, which existing approved or late-stage gene therapy program is the best biological and regulatory precedent? The answer depends on multiple parameters simultaneously — whether the delivery vector reaches the affected tissue, whether the target gene fits within the vector's packaging capacity, whether the protein biology allows for cross-correction of untransduced cells, whether the patient population is likely to carry pre-existing neutralising antibodies against the vector, whether validated promoters exist for the target tissue, and whether

the natural history of the disease allows sufficient time for treatment to be administered before irreversible damage occurs, among others.

Currently, this assessment is performed manually by expert teams at academic centres and biotechnology companies, on a disease-by-disease basis. The process is time-consuming, resource-intensive, and inaccessible to small patient advocacy groups and academic researchers who may have identified a compelling scientific case for gene therapy in a specific disease but lack the specialist expertise or funding to conduct a formal feasibility analysis. No publicly available computational tool systematically integrates the full range of biologically and clinically relevant parameters required to make this assessment across the breadth of the monogenic disease landscape.

Several databases and registries catalogue approved and investigational gene therapy programs, including ClinicalTrials.gov and disease-specific regulatory or professional-society resources. Vector biology, AAV packaging, tissue tropism, and immune response have also been studied extensively [13–17,22–25]. However, these resources address isolated aspects of the feasibility question and have not been integrated into a unified scoring framework capable of ranking candidate precedents across multiple dimensions simultaneously.

The NanoGT Framework

This study describes NanoGT, a computational framework designed to systematically address the precedent identification problem for monogenic rare diseases. For any disease with an Orphanet identifier, the framework queries public biological databases to characterise the disease and its causal gene, checks source-linked molecular mechanism evidence where curated, scores all programs in a curated surrogate database across fourteen biologically and clinically relevant dimensions, and returns a ranked list of precedent matches with a composite score, confidence classification, and plain-language rationale for each dimension score.

The fourteen scoring dimensions were selected to capture the principal determinants of gene therapy transferability: gene packaging fit within the vector, tissue tropism of the delivery vector, protein class and cross-correction potential, biological pathway similarity, molecular mechanism and therapeutic-modality compatibility, inheritance pattern compatibility, regulatory approval stage of the precedent, pre-existing vector immunogenicity in the general population, the therapeutic window available before irreversible disease progression, immune privilege of the target tissue, availability of validated tissue-specific promoters, feasibility of the required delivery route, and organelle targeting feasibility. The organelle targeting dimension addresses a category error present in single-mechanism frameworks: some proteins require delivery to specific subcellular compartments (mitochondria, peroxisomes) via post-translational import signals that must be preserved in the therapeutic construct, and some genes are encoded in the mitochondrial genome rather than the nuclear genome, making standard nuclear AAV delivery inapplicable without allotopic expression engineering. Together, these fourteen dimensions address the biological, clinical, and regulatory determinants of precedent transferability in a single integrated score.

The framework was tested on a 40-disease proof-of-concept cohort comprising positive controls, benchmark diseases with known gene-therapy precedent, a native-gene oversized-cargo stress test, a mitochondrial-delivery stress test, a pilot set of prospective monogenic rare diseases, and an extended non-loss-of-function arm covering gain-of-function, dominant-negative, RNA-toxic, haploinsufficiency, and repeat-expansion mechanisms. The surrogate database currently comprises 21 approved or clinical-stage gene therapy programmes across major vector and tissue classes. This scale is sufficient for dissertation-level proof of concept, but not for clinical decision-making.

Paper Structure

The Methods section describes the system architecture, data sources, surrogate database, and scoring logic for each of the fourteen dimensions. The Results section presents the 40-disease cohort analysis, validation and benchmark behaviour, precedent clusters, and stress-test cases. The Discussion addresses what the results support, what they do not support, limitations of the current approach, and directions for future development.

Methods

1. System Overview

A computational framework was developed to systematically match untreated monogenic diseases to approved or late-stage gene therapy programs that could serve as regulatory and scientific precedents. The framework, designated NanoGT, accepts an Orphanet disease identifier as input and produces a ranked list of gene therapy precedents alongside a structured match report. The pipeline operates in three sequential stages: (1) disease and gene characterisation from public databases, (2) multi-dimensional scoring of each candidate precedent program against the query disease, and (3) report generation with per-dimension score breakdowns and plain-language rationales.

The system was implemented in Python 3 using a modular architecture. Disease metadata was retrieved from the Orphanet REST API (api.orphanet.org). Gene and protein annotations were retrieved from the UniProt REST API (rest.uniprot.org/uniprotkb). All external API responses were cached in-session to minimise redundant network requests. A static SQLite database was used to store the curated gene therapy surrogate catalog, vector properties, and generated match results. In the event of API unavailability, a manually curated static fallback dataset covering all tested diseases was used to ensure reproducibility without internet access.

2. Disease and Gene Characterisation

2.1 Disease Metadata (Orphanet)

For each query disease, the Orphanet REST API was queried using the disease's Orphanet numeric code to retrieve the canonical disease name, gene associations, and inheritance pattern. A separate endpoint was queried to obtain cross-references to the Online Mendelian Inheritance in Man (OMIM) database. HPO (Human Phenotype Ontology) terms were supplied as part of a manually curated fallback dataset constructed from Orphanet phenotypic summaries and published literature for each tested disease.

2.2 Tissue Mapping from HPO Terms

Affected tissues were derived from HPO term names using keyword mapping. Each HPO term was searched for tissue-indicative keywords: for example, "hepat", "liver", and "jaundice" mapped to liver; "brain", "cerebr", "seizure", and "neurodegenerat" mapped to CNS; "muscul" and "myopathy" mapped to muscle; "retina", "visual", and "optic" mapped to retina. This mapping produced a list of affected tissues used as input to tropism, immune privilege, promoter availability, and route of administration scoring.

2.3 Gene and Protein Characterisation (UniProt)

For each causal gene associated with the query disease, the UniProt REST API was queried using the gene symbol with a filter for human (organism ID 9606) manually reviewed Swiss-Prot entries. The following fields were retrieved: UniProt accession, protein name, amino acid sequence length, subcellular localisation, Gene Ontology (GO) terms, UniProt keyword tags, and annotated domain features. Coding sequence (CDS) length in base pairs was estimated by multiplying amino acid sequence length by three. A protein was classified as secreted if any subcellular localisation entry contained "secreted" or "extracellular". Lysosomal localisation was identified by the presence of "lysosom" in any localisation entry or in the UniProt keyword list.

3. Gene Therapy Surrogate Database

A static database of 21 approved or late-stage gene therapy programs was manually curated from published literature, regulatory approval documents, and ClinicalTrials.gov. Programs were included if they targeted a monogenic, loss-of-function rare disease using an AAV or lentiviral vector. Each entry recorded: program name, target disease, causal gene symbol, vector serotype, primary tissue target, transgene CDS length in base pairs, approval or trial stage, approval year (where applicable), protein class (secreted, lysosomal, membrane, intracellular, or secreted_lysosomal), inheritance pattern (AR or XL), and biological pathway label.

Eight delivery vectors were catalogued: AAV1, AAV2, AAV5, AAV8, AAV9, AAVrh10, AAV2/6, and lentiviral vector (LV). For each vector, the following properties were recorded: cargo limit in base pairs, tissue tropism, CNS/retinal/hepatic/muscle tropism flags, number of existing clinical programs using that serotype, and patent restriction status. The standard cargo limit for all AAV serotypes was set at 4,700 bp, consistent with the established packaging limit for single-stranded AAV. The lentiviral vector cargo limit was set at 8,000 bp, reflecting the larger capacity of lentiviral constructs used in approved haematopoietic stem cell programs.

4. Fourteen-Dimension Scoring Engine

For each query disease and causal gene pair, every program in the surrogate database was scored across fourteen independent dimensions. Raw scores were summed and normalised to a composite score out of ten, as described in Section 5. Each dimension and its scoring logic is described below.

The scoring scheme was not trained against clinical outcome data. Instead, thresholds were selected using a transparent evidence-informed heuristic approach. Where quantitative constraints were available, such as approximate vector cargo capacity and published AAV seroprevalence estimates, these were used directly. Where no validated quantitative weighting scheme exists, such as assigning relative importance to pathway similarity, promoter availability, or route feasibility, rule-based bins were chosen to preserve interpretability and allow sensitivity analysis. Scores should therefore be interpreted as relative precedent-strength estimates rather than probabilities of clinical efficacy. A separate scoring evidence audit table was maintained to distinguish directly literature-supported parameters from heuristic score bins and to record local text anchors from the reviewed PDFs.

4.1 Packaging Fit (max 2.0)

The CDS length of the query gene was compared against the cargo limit of the vector used in the precedent program. If the query gene CDS exceeded the vector cargo limit, a hard fail was returned (score 0.0) and the program was excluded from ranking without further scoring. Otherwise, the utilisation ratio (CDS / cargo limit) was calculated and scored: $\leq 30\%$ utilisation = 2.0; 31–60% = 1.5; 61–85% = 1.0; 86–100% = 0.5. For explicitly engineered same-gene micro/mini constructs, such as DMD_micro in a Duchenne muscular dystrophy precedent, the construct CDS was used for the packaging score while the report retained a note that native full-length gene replacement remained oversized. This distinction prevents an engineered micro-transgene strategy from being misclassified as ordinary full-length gene replacement.

4.2 Tissue Tropism (max 2.0)

The set of disease-affected tissues was compared against the vector's tissue tropism list and the precedent program's tissue target description. A score of 2.0 was assigned if at least one affected tissue appeared in both the vector tropism list and the precedent program's tissue target. A score of 1.5 was assigned if the precedent program's tissue target matched at least one affected tissue, or if

the vector tropism overlapped with two or more affected tissues. A score of 1.0 was assigned for a single tissue overlap with vector tropism. A score of 0.3 was assigned where no overlap was identified. Tropism was treated as an evidence-informed proxy rather than a direct prediction of human biodistribution, because AAV tropism varies by species, route, dose, promoter, and cell type.

4.3 Protein Class (max 2.0)

The protein class of the query gene was inferred from UniProt subcellular localisation and keyword annotations, then compared to the protein class of the precedent program. A critical distinction was introduced in scoring version 2 between lysosomal enzymes (soluble, secretable proteins such as ARSA and IDUA that can cross-correct via the mannose-6-phosphate receptor pathway) and lysosomal membrane proteins (channels and transporters such as MCOLN1/mucolipin-1 and SLC17A5/sialin that are anchored in the lysosomal membrane and cannot be secreted or taken up by neighbouring cells). A score of 2.0 was assigned for an exact match between two soluble lysosomal enzymes or two secreted proteins. A score of 1.5 was assigned for a match between two intracellular or two membrane proteins. A score of 1.0 was assigned where the query gene encoded a lysosomal membrane protein matched against a lysosomal enzyme precedent (lysosomal pathway shared, but the cross-correction mechanism inapplicable), or where the query gene had some secreted or extracellular component in a mismatch context. A score of 0.5 was assigned for a clear class mismatch. Prior to this distinction, lysosomal membrane proteins received the same 2.0 score as lysosomal enzymes when matched against programmes such as Libmeldy, inflating composite scores for diseases such as mucopolidosis type IV (MCOLN1) and Salla disease (SLC17A5).

4.4 Mechanism and Modality Compatibility (max 2.0)

The molecular mechanism of the query disease was treated as a distinct evidence layer rather than inferred automatically from inheritance. A curated mechanism table (`data/disease_mechanisms.csv`) records, for each disease/gene pair in the 40-disease cohort, the mechanism category, a source-linked evidence summary, the preferred therapeutic modality class, and whether ordinary gene addition is compatible, conditional, uncertain, or incompatible. Compatible loss-of-function diseases received 2.0. Conditional cases, such as intracellular cell-autonomous proteins, oversized genes requiring engineered constructs, mitochondrial enzymes, or multi-system metabolic diseases, received 1.5. Uncertain cases received 1.0. An incompatible mechanism constitutes a second hard gate: the program is assigned a composite score of 0.0 and a confidence classification of "fail" and excluded from ranking entirely, analogous to the packaging hard gate in dimension 4.1. This prevents gain-of-function, dominant-negative, or RNA-toxic diseases from receiving plausible-looking scores on vector precedents that would be therapeutically counterproductive. Missing evidence is treated as uncertain and flagged for manual review; inheritance alone is not treated as proof of gene-addition suitability.

4.5 Inheritance Compatibility (max 1.0)

The inheritance pattern of the query disease was compared to the inheritance pattern of the precedent program. An exact match between autosomal recessive (AR) or X-linked (XL) patterns received 1.0. A shared loss-of-function pattern received 0.7. A mismatch — for example, a dominant or mitochondrial disease against a recessive precedent — received 0.3. This dimension captures the distinction between loss-of-function diseases (amenable to gene replacement) and gain-of-function or dominant negative diseases (which would require silencing or editing strategies not represented in the current database).

4.6 Biological Pathway Similarity (max 2.0)

The biological pathway of the query disease was inferred from gene keywords, GO terms, subcellular localisation, and HPO terms. Pathways considered included lysosomal storage, coagulation, motor neuron, myopathy, retinal visual cycle, retinal phototransduction, mitochondrial complex, amino acid metabolism, urea cycle, and lipid metabolism. A score of 2.0 was assigned for an exact pathway match. A score of 1.5 was assigned for a related pathway, defined by pre-specified groupings of mechanistically adjacent pathways (for example, amino acid metabolism and urea cycle). A score of 0.5 was assigned for a clearly different pathway. Where the pathway could not be inferred, a neutral score of 1.0 was assigned.

4.7 Regulatory Approval Weight (max 1.0)

The regulatory stage of the precedent program was used to weight the strength of its precedent value. Scores were assigned as follows: approved = 1.0; withdrawn (previously approved) = 0.7; Phase 3 = 0.8; Phase 2/3 = 0.7; Phase 2 = 0.6; Phase 1/2 = 0.5; Phase 1 = 0.4. This dimension was interpreted as a proxy for accumulated development evidence, including manufacturing, toxicology, clinical design, endpoint, and regulator-interaction experience, not as a guarantee of efficacy.

4.8 Vector Immunogenicity (max 2.0)

Pre-existing neutralising antibodies (NAbs) against AAV capsids in the general population are a well-established barrier to treatment eligibility, as patients with NAb titres above defined thresholds are typically excluded from many AAV clinical trials. Population seroprevalence estimates for each vector serotype were sourced from published studies and reviews of AAV humoral immunity [22–25] and used to score immunogenicity risk. Scores were assigned based on approximate seroprevalence: <10% = 2.0; 10–19% = 1.5; 20–39% = 1.0; ≥40% = 0.5. Lentiviral vectors were assigned minimum seroprevalence (~2%) as they are immunologically distinct from AAV serotypes. This dimension captured only vector-level pre-existing humoral immunity; it did not explicitly model complement activation, dose-related toxicity, anti-transgene immunity, T-cell responses, immunosuppression protocols, or redosing feasibility.

4.9 Therapeutic Window (max 2.0)

The therapeutic window dimension assessed whether gene therapy could feasibly be administered before irreversible cellular damage occurs. This was inferred from HPO terms and disease name using keyword analysis. Diseases with adult or slowly progressive onset received 2.0. Progressive childhood-onset diseases with neurodegeneration received 1.5. Diseases with early childhood onset where newborn screening integration would substantially improve outcomes received 1.2. Neonatal-onset diseases requiring delivery within weeks of birth received 0.8. Congenital or rapidly fatal early-onset diseases requiring in utero or immediate neonatal delivery received 0.5. This dimension does not penalise the match per se but flags where clinical trial design is substantially more complex.

4.10 Cross-Correction Potential (max 1.0)

Cross-correction describes the ability of a transduced cell to secrete the therapeutic protein and thereby correct adjacent untransduced cells. This mechanism amplifies the effective reach of gene delivery without requiring every target cell to receive the vector individually, enabling lower doses and broader tissue coverage. Secreted proteins and secreted lysosomal enzymes (taken up by neighbouring cells via the mannose-6-phosphate receptor pathway) received 1.0. Non-secreted soluble lysosomal enzymes with partial cross-correction capacity received 0.8. Lysosomal membrane proteins (channels and transporters, e.g. MCOLN1, SLC17A5) received 0.0: these proteins are anchored in the lysosomal membrane and cannot be released as soluble cargo, so every target cell must individually receive the vector — no bystander rescue is possible. Intracellular

and non-lysosomal membrane-bound proteins, for which no cross-correction is possible, received 0.2. This correction was introduced in scoring version 2; prior to it, lysosomal membrane proteins were erroneously assigned 0.8 as if they were soluble lysosomal enzymes.

4.11 Immune Privilege of Target Tissue (max 1.0)

Certain tissues are immunologically privileged: they are partially sequestered from systemic immune surveillance by physical barriers or local immunosuppressive signals. High immune privilege is associated with reduced risk of cytotoxic T-lymphocyte-mediated clearance of transduced cells and more durable long-term transgene expression. Scores were assigned by tissue: retina = 1.0 (blood-retinal barrier, FasL expression, local TGF- β 2); CNS = 0.9 (blood-brain barrier); liver = 0.8 (tolerogenic microenvironment: Kupffer cells, IL-10, PD-L1); muscle and heart = 0.6; kidney = 0.5; haematopoietic tissue = 0.3.

4.12 Promoter Availability (max 1.0)

The availability of validated, tissue-specific promoters was scored based on the disease's primary target tissue. Validated promoters are essential for restricting transgene expression to the intended cell type and for ensuring therapeutic expression levels. Liver and retinal targets received 1.0, reflecting the availability of multiple clinically validated promoters used in approved programs (ApoE/hAAT, TBG, and transthyretin for liver; VMD2, GRK1, and CRX for retina). CNS and muscle targets received 0.8 (Synapsin-1 and CaMKII for CNS; MHCK7 and CK8 for muscle). Haematopoietic targets received 0.7. Cardiac targets received 0.6. Renal targets received 0.4, reflecting the near-absence of validated kidney-specific promoters in clinical programs.

4.13 Route of Administration Feasibility (max 1.0)

The feasibility of delivering the vector to the target tissue via an established clinical route was scored based on the disease's affected tissues. Hepatic delivery via intravenous infusion received 1.0, reflecting its use in all approved hepatic gene therapy programs. Muscle delivery via intravenous or intramuscular injection received 0.9. Haematopoietic delivery via ex vivo haematopoietic stem cell modification and reinfusion received 0.9. Retinal delivery via subretinal or intravitreal injection received 0.8, reflecting the established surgical approach used in Luxturna and GS010. CNS delivery via intrathecal or intracerebroventricular injection received 0.7, reflecting higher procedural risk. Cardiac delivery received 0.6. Renal delivery received 0.4, reflecting the absence of an established kidney-specific delivery route in approved programs.

4.14 Organelle Targeting Feasibility (max 1.0)

This dimension assesses whether standard nuclear AAV delivery can produce a functional protein at the correct subcellular compartment for the query disease. It was introduced in scoring version 2 to address a category error in the original implementation: certain diseases were receiving scores based on comparison to nuclear gene-addition programs even when the causal gene requires a fundamentally different delivery logic.

Mitochondrial DNA-encoded genes (MT-* genes, identified by HGNC symbol convention) received a score of 0.0. These genes reside in the mitochondrial genome, use a non-standard genetic code, and are translated by mitochondrial ribosomes inside the organelle. Standard nuclear AAV delivery is inapplicable without allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial mitochondrial targeting sequence. This strategy is disease-specific and not represented in the current surrogate catalogue. Leber hereditary optic neuropathy (MT-ND4) is the primary example; GS010/Lumevoq (lenadogene nolpharvovec) uses this allotopic approach as a real-world precedent, though its EMA marketing authorisation application was voluntarily withdrawn in April 2023 after Phase 3 trials did not demonstrate statistically significant efficacy versus sham

injection.

Nuclear-encoded mitochondrial matrix or inner-membrane proteins received a score of 0.5. Nuclear AAV delivery is theoretically feasible, as the gene resides in the nuclear genome, but the N-terminal mitochondrial targeting sequence (MTS) in the therapeutic construct must be intact and functional for correct post-translational import. MTS integrity is not validated by any other scoring dimension and requires disease-specific experimental confirmation. Methylmalonic acidemia (MUT) is the primary example.

Other nuclear-encoded mitochondrial proteins (outer membrane, intermembrane space) received 0.6. Peroxisomal proteins received 0.7, reflecting that AAV delivery is feasible but the peroxisomal targeting signal (PTS1/PTS2) must be preserved in the construct. Proteins with standard subcellular localisation (cytoplasmic, nuclear, secreted, lysosomal, endoplasmic reticulum, plasma membrane) received 1.0, as nuclear AAV delivery directly produces functional protein at the correct location.

5. Score Normalisation and Confidence Classification

Raw scores across all fourteen dimensions were summed to produce a raw composite with a theoretical maximum of 21.0. This was normalised to a score out of ten using the formula:

Composite score = (raw sum / 21.0) × 10

Programs were classified into confidence tiers based on the normalised composite score: high confidence (≥ 7.5), medium confidence (5.0–7.49), and low confidence (< 5.0). Programs that failed the packaging gate (dimension 4.1) received a composite score of 0.0 and a classification of "fail", indicating physical incompatibility between the query gene and the candidate vector.

6. Cohort Design

The current dissertation analysis used a 40-disease proof-of-concept cohort stored in `data/disease_cohort_40.csv`, with source-linked mechanism evidence stored separately in `data/disease_mechanisms.csv`. The cohort was intentionally structured to test more than one success mode: positive controls with approved precedents, benchmark diseases with known gene-therapy precedent, an oversized native-cargo stress test, a mitochondrial-delivery stress test, and a broader pilot cohort of monogenic diseases spanning liver, CNS, retina, muscle, heart, and haematopoietic involvement. Each cohort row records the Orphanet identifier, disease name, causal gene, inheritance pattern, affected tissues, prevalence class, OMIM cross-reference, cohort role, source URL, and fact-check status. Each mechanism row records the disease/gene mechanism category, gene-addition compatibility, preferred modality class, evidence summary, evidence URL, citation label, and evidence verification status.

The cohort should be interpreted as a reproducible dissertation test set rather than a statistically representative sample of all rare diseases. Its purpose is to demonstrate whether the scoring framework produces interpretable precedent rankings and useful failure modes across diverse disease classes.

7. Output and Report Generation

Results were sorted in descending order of composite score for each disease query. A structured Markdown report was generated containing: source-linked disease mechanism evidence; a ranked summary table of top matches with composite score and confidence tier; a per-match breakdown table showing the score for each of the fourteen dimensions; plain-language rationale notes for each dimension; manual review flags; and a log of packaging-excluded programs. Reports were saved to

a standardised output directory using filenames incorporating the Orphanet ID and disease name. A cross-disease summary report was generated to allow comparison of top matches, mechanism categories, gene-addition compatibility labels, and score distributions across all queried diseases.

Each report includes a review flags section that captures translational caveats not reflected in composite rank alone. Version 2 of the framework expanded the flag categories to include: lysosomal membrane protein flags (identifying diseases where the M6P cross-correction mechanism of haematopoietic stem cell precedent programmes is not applicable); mitochondrial DNA gene flags (identifying MT-* genes requiring allotopic expression rather than standard nuclear gene addition); mitochondrial matrix enzyme flags (identifying nuclear-encoded mitochondrial proteins where intact mitochondrial targeting sequence function must be validated); multi-subunit enzyme complex flags (identifying diseases where a single subunit delivered by the vector may not reconstitute full enzymatic activity without co-delivery of other subunits); unresolved disease biology flags (identifying genes whose molecular function is not fully characterised in the curated evidence); and disease heterogeneity flags (identifying diseases such as Gaucher disease and Fabry disease where neuronopathic and non-neuronopathic subtypes require fundamentally different therapeutic strategies and scores should be interpreted subtype-specifically). These flags are designed to be read alongside composite scores and not suppressed: a high-ranking match can legitimately carry several flags requiring review before it is used as a development template.

Results

1. Cohort Overview

The NanoGT framework was applied to a 40-disease monogenic rare-disease cohort assembled from Orphanet identifiers and manually curated gene/tissue metadata. The cohort was deliberately mixed rather than restricted to easy discovery cases. It contained:

- 2 positive controls with approved gene-therapy precedents.
- 2 benchmark diseases with known gene-therapy precedent.
- 1 oversized-cargo stress test.
- 1 mitochondrial-delivery stress test.
- 24 pilot discovery or extension diseases.
- 6 haploinsufficiency diseases (Rett syndrome, Dravet syndrome, CDKL5 deficiency, GATA2 deficiency, CHARGE syndrome, Neurofibromatosis type 1).
- 2 repeat-expansion/silencing diseases (Fragile X syndrome, Friedreich ataxia).
- 1 genomic imprinting disease (Angelman syndrome).
- 1 two-hit haploinsufficiency disease (Tuberous sclerosis complex).

Each query disease was scored against the curated 21-program surrogate catalogue. The algorithm uses fourteen dimensions: packaging fit, tissue tropism, protein class, pathway similarity, mechanism/modality compatibility, inheritance compatibility, regulatory approval stage, vector immunogenicity, therapeutic window, cross-correction potential, immune privilege, promoter availability, route-of-administration feasibility, and organelle targeting feasibility. Raw scores have a maximum of 21 and are normalised to a score out of 10.

2. Cross-Disease Score Distribution

All 40 diseases received at least some result from the framework — there are no mechanism hard-fails in this cohort because the non-LOF arm was deliberately designed from conditional diseases (haploinsufficiency, repeat-expansion silencing, genomic imprinting) where gene addition remains a biologically rational strategy. One disease, Neurofibromatosis type 1 (NF1, 8,451 bp CDS), received a packaging hard-fail: its coding sequence exceeds the capacity of every vector in the current catalogue, so no precedent can be ranked. All other 39 diseases received at least one scored precedent.

Among the 39 scored diseases, Duchenne muscular dystrophy remains an expected stress-test case: full-length native DMD fails the ordinary single-AAV packaging gate, but the catalogue contains SRP-9001 as an engineered micro-dystrophin precedent, so a score is still returned. The framework separates ordinary full-length gene replacement failure from engineered micro/mini-transgene strategy scoring.

Among the 39 scored diseases, 32 top matches were classified as high confidence and 7 as medium confidence. Composite scores ranged from 5.9/10 to 9.9/10. These values should be interpreted as relative precedent strength, not predicted clinical efficacy.

Table 1. Forty-disease NanoGT cohort results

Cohort role	Disease	ORPHA	Gene	Mechanism	Gene-addition (bp)	Top precedent	Vector	Score	Confidence	
positive_control	hemophilia	BORPHA:306	F9	loss_of_function	compatible	1383	Hemgenix	AAV5	9.9/10	high
benchmark	with known gene-therapy precedent	ORPHA:105	RPE65	loss_of_function	compatible	1599	CPCB-RPE1	AAV8	7.6/10	high

Cohort role	Disease	ORPHA	Gene	Mechanism	Gene-addition	ODB (bp)	Top precede	Vector	Score	Confidence
benchmark_variant	Severe combined immunodeficiency due to adenosine deaminase deficiency	ORPHA:127	ADA	loss_of_function	compatible	1092	Strimvelis	LV	8.1/10	high
positive_control	Spinal Muscular Atrophy	ORPHA:10	SMN1	loss_of_function	compatible	891	OAV101-IT	AAV9	8.3/10	high
oversized_caducress	Duchenne muscular dystrophy	ORPHA:198	DMD	loss_of_function	conditional	11055	SRP-9001	AAV9	7.6/10	high
mitochondrial	Leber hereditary optic neuropathy	ORPHA:104	MTND4	mitochondrial	loss_of_function	1877	CPCB-RPE1	AAV8	6.4/10	medium
pilot_cohort	Achromatopsia	ORPHA:49382	NGB3	loss_of_function	compatible	2427	CPCB-RPE1	AAV8	7.5/10	medium
pilot_cohort	Alpha-mannosidosis	ORPHA:61	MAN2B1	loss_of_function	compatible	3033	Libmeldy	LV	8.8/10	high
pilot_cohort	Choroideremia	ORPHA:180	CHM	loss_of_function	compatible	1962	Luxturna	AAV2	8.0/10	high
pilot_cohort	Crigler-Najjar syndrome	ORPHA:106	UGT1A1	loss_of_function	compatible	1596	Hemgenix	AAV5	7.2/10	medium
pilot_cohort	Fabry disease	ORPHA:324	GLA	loss_of_function	compatible	1290	Libmeldy	LV	8.9/10	high
pilot_cohort	Gaucher disease	ORPHA:355	GBA	loss_of_function	conditional	1491	Libmeldy	LV	9.0/10	high
pilot_cohort	Glycogen storage disease type II	ORPHA:725	PFKP	loss_of_function	compatible	1071	SRP-9001	AAV9	7.8/10	high
pilot_cohort	Kohlschütter-Janz syndrome	ORPHA:196	ROGDI	loss_of_function	conditional	861	OAV101-IT	AAV9	7.7/10	high
pilot_cohort	Krabbe disease	ORPHA:487	GALC	loss_of_function	conditional	2055	Libmeldy	LV	9.0/10	high
pilot_cohort	Maple syrup urine disease	ORPHA:54	BCKDHA	loss_of_function	conditional	1335	BMN 307	AAV5	8.0/10	high
pilot_cohort	Metachromatic leukodystrophy	ORPHA:512	ARSA	loss_of_function	compatible	1521	Libmeldy	LV	9.2/10	high
pilot_cohort	Mucopolysaccharidosis type IX	ORPHA:578	MCOLN1	loss_of_function	conditional	1740	Skysona	LV	8.3/10	high
pilot_cohort	Mucopolysaccharidosis type I	ORPHA:519	IDA	loss_of_function	compatible	1962	Libmeldy	LV	9.5/10	high
pilot_cohort	Mucopolysaccharidosis type II	ORPHA:549	IDB	loss_of_function	compatible	1650	Libmeldy	LV	9.3/10	high
pilot_cohort	Mucopolysaccharidosis type III (Sanfilippo syndrome)	ORPHA:592	SGSH	loss_of_function	compatible	1674	Libmeldy	LV	9.2/10	high
pilot_cohort	Ornithine transcarbamoylase deficiency	ORPHA:66	OTC	loss_of_function	compatible	1065	BMN 307	AAV5	8.1/10	high
pilot_cohort	Phenylketonuria	ORPHA:716	PAH	loss_of_function	compatible	1353	BMN 307	AAV5	8.0/10	high
pilot_cohort	Pompe disease	ORPHA:365	GAA	loss_of_function	conditional	2856	Libmeldy	LV	8.4/10	high
pilot_cohort	Salla disease	ORPHA:309	SLC17A5	loss_of_function	conditional	1485	Skysona	LV	8.1/10	high
pilot_cohort	Vitamin B12 deficiency (Methylmalonic acidemia)	ORPHA:25	MMH	loss_of_function	conditional	2250	BMN 307	AAV5	7.7/10	high
pilot_cohort	Wiskott-Aldrich syndrome	ORPHA:906	WAS	loss_of_function	compatible	1506	Strimvelis	LV	8.0/10	high
pilot_cohort	X-linked adrenoleukodystrophy	ORPHA:43	ABCD1	loss_of_function	conditional	2235	Skysona	LV	8.2/10	high
pilot_cohort	X-linked myotubular myopathy	ORPHA:159	MTM1	loss_of_function	compatible	1878	AT132	AAV8	7.7/10	high
pilot_cohort	X-linked retinoschisis	ORPHA:792	RS1	loss_of_function	compatible	672	Luxturna	AAV2	7.9/10	high
non_lof_haploinsufficiency	Residual syndrome	ORPHA:778	MECP2	haploinsufficiency	conditional	1461	Skysona	LV	7.4/10	medium
non_lof_haploinsufficiency	Unsatisfactory	ORPHA:1306	SCN1A	haploinsufficiency	conditional	6027	Skysona	LV	7.1/10	medium
non_lof_repeat_expansion	Fragile X syndrome	ORPHA:908	FMR1	repeat_expansion	conditional	1899	Skysona	LV	7.0/10	medium
non_lof_imprinting	Angelman syndrome	ORPHA:72	UBE3A	genomic_imprinting	conditional	2598	OAV101-IT	AAV9	7.1/10	medium
non_lof_haploinsufficiency	Oncofibrinogen deficiency	ORPHA:1063	CKL5	haploinsufficiency	conditional	3093	Skysona	LV	7.2/10	medium
non_lof_haploinsufficiency	Oncofibrinogen deficiency	ORPHA:2477	TA2	haploinsufficiency	conditional	1443	Strimvelis	LV	7.6/10	high
non_lof_haploinsufficiency	Oncofibrinogen deficiency	ORPHA:138	CHD7	haploinsufficiency	conditional	7950	Libmeldy	LV	5.9/10	medium
non_lof_haploinsufficiency	Unsatisfactory	ORPHA:636	NF1	haploinsufficiency	conditional	8451	—	—	—	packaging_hard_fail
non_lof_haploinsufficiency	Unsatisfactory	ORPHA:805	ESC2	haploinsufficiency	conditional	5694	Libmeldy	LV	6.1/10	medium

Cohort role	Disease	ORPHA	Gene	Mechanism	Gene-addition	DB (bp)	Top precedent	Vector	Score	Confidence
non_lof_repeat_expansion	Friedreich's ataxia	ORPHA:95	FXN	repeat_expansion	codon_tolerant	633	OAV101-IT	AAV9	7.9/10	high

3. Top-Precedent Clusters

The most important result is not any single disease score, but the emergence of biologically interpretable precedent clusters.

3.1 Lentiviral / Libmeldy-dominated lysosomal and leukodystrophy cluster

Libmeldy was the top-ranked precedent for 9 diseases: alpha-mannosidosis, Fabry disease, Gaucher disease, Krabbe disease, metachromatic leukodystrophy, mucopolysaccharidosis types I/II/IIIA, and Pompe disease. This cluster is plausible because ex vivo lentiviral haematopoietic stem-cell gene therapy has a strong precedent base for lysosomal and neurodegenerative metabolic disorders, particularly where cross-correction by enzyme secretion or haematopoietic-derived CNS engraftment may contribute to benefit.

Importantly, mucopolipidosis type IV and Salla disease — which share the CNS/lysosomal surface label — did not have Libmeldy as their top precedent in the final scoring run. Both ranked Skysona (LV) first. This is a meaningful result: MCOLN1 (ML-IV) encodes a lysosomal membrane channel and SLC17A5 (Salla) encodes a lysosomal membrane transporter; neither protein can be secreted or taken up by adjacent cells via the mannose-6-phosphate receptor pathway. The Skysona precedent, which uses a lentiviral vector for CNS-directed ex vivo haematopoietic correction in a leukodystrophy setting, captures the CNS/lentiviral class without implying classical enzyme cross-correction. This distinction — and the limitation that the current algorithm does not fully resolve it — must be stated clearly in the dissertation.

3.2 Hepatic metabolic / BMN 307 cluster

BMN 307 was top-ranked for 4 diseases: maple syrup urine disease, ornithine transcarbamylase deficiency, phenylketonuria, and vitamin B12-unresponsive methylmalonic acidemia. These diseases share hepatic and metabolic features, small-to-moderate transgene sizes, and a plausible liver-directed AAV development logic. The framework therefore identifies BMN 307 as a useful platform-style precedent for liver metabolic gene-addition programmes.

The interpretation should remain cautious. These diseases differ in subcellular localisation, metabolic pathway, newborn-screening relevance, and the extent to which liver correction would address neurological injury. For the dissertation, the BMN 307 cluster is best presented as a hypothesis-generating prioritisation result, not as a statement that the same vector/programme can be directly reused.

3.3 Retinal precedent cluster

Retinal diseases clustered around Luxturna and CPCB-RPE1. Choroideremia and X-linked retinoschisis were high-confidence retinal matches, while Leber congenital amaurosis, Leber hereditary optic neuropathy, and achromatopsia were medium-confidence matches. This spread is biologically sensible: the eye is accessible and immune-privileged, but gene-specific biology, affected retinal cell type, mitochondrial inheritance, and programme stage all affect transferability.

3.4 Positive controls and benchmark behaviour

The strongest positive control was haemophilia B: Hemgenix ranked first with a 9.9/10 score. Spinal muscular atrophy also returned an AAV9/Zolgensma/OAV101-style precedent at high confidence. ADA-SCID ranked with Strimvelis, correctly recovering an ex vivo haematopoietic precedent. These

results support the internal validity of the scoring system at a proof-of-concept level.

Leber congenital amaurosis did not rank Luxturna first in the current run; CPCB-RPE1 ranked first at 7.5/10, with Luxturna second in the generated disease report. This is not a trivial detail to hide. It indicates that the current scoring formula may over-weight vector/tissue/programme-stage similarity relative to exact disease identity for some retinal cases. In the dissertation, this should be discussed as a calibration issue and a motivation for either exact-target bonus scoring or a clearer validation metric.

4. Medium-Confidence and Stress-Test Results

Medium-confidence top matches among scored diseases were:

- Leber hereditary optic neuropathy (MT-ND4): CPCB-RPE1 / AAV8 at 6.4/10
- Achromatopsia (CNGB3): CPCB-RPE1 / AAV8 at 7.5/10
- Crigler-Najjar syndrome type I (UGT1A1): Hemgenix / AAV5 at 7.2/10
- CHARGE syndrome (CHD7): Libmeldy / LV at 5.9/10
- Tuberous sclerosis complex (TSC2): Libmeldy / LV at 6.1/10

These cases are useful because they show where the framework becomes uncertain. They include retinal diseases where several plausible precedents compete, a mitochondrial disease where ordinary nuclear gene addition is mechanistically uncertain, a liver disease where pathway and protein-class similarity are imperfect, and two haploinsufficiency diseases where conditional gene-addition logic and large transgene size both constrain tractability.

Duchenne muscular dystrophy illustrates the distinction between native-gene replacement and engineered-cargo precedent. Full-length DMD fails ordinary single-AAV packaging, but SRP-9001 can be scored as a micro-dystrophin strategy. The correct interpretation is not that DMD is unsuitable for gene therapy, but that DMD requires engineered microgene, dual-vector, exon-skipping, or editing-based logic rather than ordinary full-length gene addition.

5. Non-LOF Mechanism Arm

The non-LOF arm contains 10 diseases spanning haploinsufficiency, repeat-expansion silencing, genomic imprinting, and two-hit haploinsufficiency mechanisms. Unlike purely incompatible diseases (gain-of-function, dominant-negative), all 10 were selected because gene addition is a biologically rational strategy — they are genuine candidates for gene therapy, not diseases where adding a copy makes things worse. The key dissertation question is whether the scoring framework produces interpretable, biologically appropriate results for mechanisms that differ from classical loss-of-function.

The six haploinsufficiency CNS diseases (Rett, Dravet, CDKL5, GATA2, CHARGE, plus TSC as a two-hit variant) all returned medium-to-high confidence matches. The four CNS haploinsufficiency diseases with small-to-moderate gene size (Rett/MECP2, Dravet/SCN1A, CDKL5/CDKL5, Angelman/UBE3A) returned scores of 7.1–7.4, with Skysona (LV) or OAV101-IT (AAV9) as top precedents. These scores are interpretable: the diseases share CNS involvement, small-to-moderate gene size, and a case for ex vivo or in vivo CNS lentiviral delivery. The Skysona precedent is appropriate because it is a CNS-targeted lentiviral program, even though the precise mechanism differs.

GATA2 deficiency scored 7.6/10 (high confidence), with Strimvelis (LV) as top precedent. This is the most mechanistically coherent non-LOF result: GATA2 haploinsufficiency causes haematopoietic failure and the ex vivo HSC gene-addition logic of Strimvelis maps cleanly to GATA2 biology.

The two large haploinsufficiency/two-hit diseases — CHARGE syndrome (CHD7, 7,950 bp) and Tuberous sclerosis complex (TSC2, 5,694 bp) — scored 5.9 and 6.1 (medium), correctly reflecting that large CDS size and the two-hit nature of TSC reduce precedent transferability.

Neurofibromatosis type 1 (NF1, 8,451 bp) received a packaging hard-fail: its CDS exceeds the capacity of every vector in the current catalogue. This is the correct result — NF1 is not a disease where standard single-vector gene addition is tractable, and the empty result is a valid scientific finding about packaging constraints.

The two repeat-expansion diseases behaved differently and this contrast is instructive. Friedreich ataxia (FXN, 633 bp) scored 7.9/10 high confidence with OAV101-IT (AAV9) as top precedent. This is correct because the frataxin protein coding sequence is structurally normal — the GAA repeat silences the endogenous gene, but FXN cDNA addition bypasses the silenced allele. The tool treats it similarly to a CNS/cardiac LOF disease, which is the right biological analogy. Fragile X syndrome (FMR1, 1,908 bp) scored 7.0/10 medium, with Skysona as top precedent. FMR1 cDNA addition is also clinically pursued for the same silencing-bypass logic, and the medium score reflects appropriate caution: FMRP has synaptic dosage-sensitivity that makes titrating expression more complex than frataxin.

6. Summary of Findings

The 40-disease analysis supports four dissertation-level claims:

- NanoGT can recover obvious clinical precedents for positive controls and benchmark diseases.
- Cross-disease clustering produces biologically plausible precedent groups, especially for lentiviral/HSC lysosomal disease, liver metabolic AAV programmes, and retinal AAV programmes.
- The framework usefully exposes hard limitations: oversized native cargo (DMD, NF1 packaging failures), mitochondrial allotopic-expression requirements (LHON), and the distinction between secreted lysosomal enzymes and lysosomal membrane proteins.
- The non-LOF arm produces interpretable, biologically coherent results for conditional mechanisms (haploinsufficiency, repeat-expansion silencing, genomic imprinting), demonstrating that the framework extends meaningfully beyond classical LOF.

The results should be framed as a reproducible prioritisation and precedent-mapping framework, not as a validated therapeutic recommendation system.

Discussion

1. Principal Findings

This dissertation developed and tested NanoGT, a computational framework for matching monogenic rare diseases to existing gene-therapy precedents. The current implementation was applied to a 40-disease cohort and produced interpretable scored results for 39 diseases. The one exception, Neurofibromatosis type 1, received a packaging hard-fail because NF1 (8,451 bp) exceeds the capacity of every vector in the catalogue — a biologically informative result in its own right. Duchenne muscular dystrophy remains biologically informative: the native DMD coding sequence fails the ordinary single-vector packaging gate, while SRP-9001 is surfaced only as an engineered micro-dystrophin precedent. This distinction prevents microgene strategies from being mistaken for ordinary full-length gene replacement.

The central finding is that the framework does not return random high scores. Instead, it organises diseases into recognisable precedent clusters: lentiviral haematopoietic stem-cell therapy for lysosomal/leukodystrophy-like disorders, liver-directed AAV programmes for metabolic diseases, and retinal AAV programmes for inherited retinal disorders. These clusters are plausible given the underlying biology, vector delivery history, and regulatory precedent landscape.

2. Internal Validation

The positive-control behaviour supports the proof-of-concept value of the scoring system. Haemophilia B ranked Hemgenix as the top precedent with the highest score in the cohort. ADA-SCID mapped to Strimvelis, correctly identifying an ex vivo haematopoietic approach. Spinal muscular atrophy mapped to the AAV9/Zolgensma/OAV101 precedent space. These results show that the framework can recover obvious known precedents when they exist in the catalogue.

However, validation is not perfect. Leber congenital amaurosis did not return Luxturna as the single top-ranked result in the current summary, despite Luxturna being the canonical approved RPE65 therapy. CPCB-RPE1 ranked first, with Luxturna close behind. This is an important calibration issue rather than something to conceal. The current scoring formula can over-weight shared vector/tissue/programme-stage features relative to exact disease identity. In the final dissertation, validation should therefore be reported using top-k recovery and qualitative biological plausibility, not only top-1 accuracy.

3. Biological and Translational Interpretation

The dominance of Libmeldy as a precedent for lysosomal and leukodystrophy-like disorders is one of the most coherent outputs of the 40-disease run. Ex vivo lentiviral haematopoietic stem-cell therapy is a strong development precedent for diseases where corrected cells can engraft, migrate, or provide cross-correcting enzyme. This supports the idea that NanoGT can surface regulatory and delivery precedents that may not be obvious from disease name alone.

At the same time, the Libmeldy cluster demonstrates why the tool must remain interpretive rather than automatic. Some lysosomal labels refer to secreted enzymes with cross-correction potential; others refer to lysosomal membrane channels or transporters that require cell-autonomous correction. Treating those as equivalent would overstate the translational readiness of several matches. The dissertation should use this cluster as both a success and a limitation: the framework identifies the right neighbourhood, but expert biological interpretation is needed before moving from neighbourhood to development plan.

The hepatic metabolic cluster around BMN 307 is similarly useful but not definitive. Liver-directed AAV is a plausible precedent class for several metabolic diseases, but each disease differs in newborn-screening availability, subcellular localisation, disease reversibility, toxic metabolite kinetics, and whether liver correction alone is sufficient. The retinal cluster is clinically plausible because the eye has established delivery routes and immune privilege, but exact target cell type and disease mechanism remain critical. For this reason, the current implementation now includes a source-linked mechanism/modality layer rather than relying on inheritance alone as a proxy for loss of function.

4. Limitations

4.1 Catalogue coverage

The current surrogate catalogue contains 21 gene-therapy programmes. This gives useful coverage of AAV and lentiviral precedents, but it is not a comprehensive clinical-trial landscape. The catalogue is biased toward better-known approved or late-stage programmes and under-represents kidney, cardiac, peripheral nervous system, mitochondrial, dominant-negative, RNA, editing, and non-viral delivery approaches. Any disease whose best precedent lies outside the catalogue may receive a misleadingly weak or indirect match.

4.2 Manual disease facts and source verification

The 40-disease CSV is useful for reproducibility, but every row is still marked as requiring fact-checking. Before final submission, the disease name, ORPHA identifier, causal gene, inheritance, tissues, prevalence class, and OMIM cross-reference must be checked against Orphanet, OMIM, UniProt, and at least one disease-specific review or primary paper. This is now a dissertation-critical work item, not a cosmetic clean-up task.

The same applies to the mechanism evidence table. `data/disease_mechanisms.csv` prevents the framework from pretending that inheritance equals molecular mechanism, but each source link and compatibility label should be treated as an auditable claim. Where the mechanism is poorly understood or the evidence is indirect, the correct label is unknown, uncertain, or conditional rather than a forced loss-of-function assignment.

4.3 Heuristic pathway and therapeutic-window inference

Pathway similarity and therapeutic window are currently inferred by keyword-style heuristics. This is acceptable for a proof-of-concept dissertation if stated honestly, but it is not equivalent to curated pathway modelling or formal natural-history extraction. Future work should integrate Reactome/KEGG/GO identifiers and structured natural-history fields from Orphanet or OMIM.

4.4 Cross-correction and protein localisation

The current cross-correction score is too coarse. It does not fully separate secreted proteins, secreted lysosomal enzymes, non-secreted lysosomal enzymes, lysosomal membrane proteins, cytosolic proteins, nuclear proteins, and mitochondrial proteins. This matters because vector spread and required transduction fraction differ dramatically across these categories.

4.5 Scope restriction to gene addition

NanoGT v0.1 is fundamentally a gene-addition precedent matcher with limited engineered-cargo handling. It is not designed for dominant-negative disease, toxic gain-of-function disease, repeat-expansion disease, RNA knockdown, antisense therapy, base editing, prime editing, dual-AAV reconstruction, or mitochondrial replacement strategies. DMD and LHON illustrate this boundary: DMD requires explicit microgene/engineering logic, while LHON depends on specialised

mitochondrial or allotopic-expression assumptions rather than ordinary nuclear gene addition.

4.6 Score calibration

Composite scores are heuristic relative scores, not calibrated probabilities of success. The framework has not yet been trained or validated against a large set of clinical outcomes. Therefore, scores should be used for prioritisation and explanation, not for go/no-go decisions.

5. Implications for the Dissertation

The strongest dissertation framing is not "NanoGT solves rare-disease gene therapy selection." The defensible framing is:

- NanoGT is a reproducible proof-of-concept framework for precedent mapping.
- It integrates biological, vector, clinical, and regulatory dimensions in one scoring system across fourteen dimensions.
- It can recover several known precedents and generate interpretable disease clusters across a 40-disease cohort.
- It explicitly exposes failure modes: oversized cargo, mitochondrial allotopic requirements, and NF1's packaging hard-fail; and it produces interpretable conditional-confidence results for the non-LOF arm.

This is enough for a strong dissertation if the report is honest, well-referenced, and clear about limitations.

6. Future Work

The next version should prioritise: expansion and versioning of the surrogate catalogue; formal source provenance for every disease fact; exact-target bonus scoring for validation cases; better cross-correction/protein-localisation classes; Reactome/KEGG/GO pathway mapping; natural-history extraction; top-k validation metrics; and support for non-standard therapeutic modalities such as micro-gene, dual-vector, editing, RNA, and mitochondrial strategies.

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Lysosomal Storage Diseases and Gene Therapy

- Biffi A, Montini E, Lorioli L, Cesani M, Fumagalli F, Plati T, et al. Lentiviral hematopoietic stem cell gene therapy benefits metachromatic leukodystrophy. *Science*. 2013;341(6148):1233158. doi:10.1126/science.1233158

<!-- Lentiviral ex vivo HSC for lysosomal storage disease — precedent for AVR-RD-01 approach -->

- Tardieu M, Zerah M, Gougeon ML, Ausseil J, de Bournonville S, Husson B, et al. Intracerebral gene therapy in children with mucopolysaccharidosis type IIIB syndrome: an uncontrolled phase 1/2 clinical trial. *Lancet Neurol*. 2017;16(9):712–720. doi:10.1016/S1474-4422(17)30169-2

<!-- MPS IIIB (Sanfilippo B) intracranial gene therapy — direct precedent for ABO-101 -->

- Hinderer C, Katz N, Buza EL, Dyer C, Goode T, Bell P, et al. Severe toxicity in nonhuman primates and piglets following high-dose intravenous administration of an adeno-associated virus vector expressing human SMN. *Hum Gene Ther*. 2018;29(3):285–298. doi:10.1089/hum.2018.015

<!-- High-dose AAV toxicity — relevant to therapeutic window and dosing considerations -->

Haematopoietic Stem Cell Gene Therapy

- Aiuti A, Biasco L, Scaramuzza S, Ferrua F, Cicalese MP, Baricordi C, et al. Lentiviral hematopoietic stem cell gene therapy in patients with Wiskott-Aldrich syndrome. *Science*. 2013;341(6148):1233151. doi:10.1126/science.1233151
- Cavazzana-Calvo M, Payen E, Negre O, Wang G, Hehir K, Fusil F, et al. Transfusion independence and HMGA2 activation after gene therapy of human β -thalassaemia. *Nature*. 2010;467(7313):318–322. doi:10.1038/nature09328

<!-- Lentiviral ex vivo HSC landmark study -->

Protein Databases and Bioinformatics Tools

- UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Res*. 2023;51(D1):D523–D531. doi:10.1093/nar/gkac1052
- The Gene Ontology Consortium. The Gene Ontology resource: enriching a GOld mine. *Nucleic Acids Res*. 2021;49(D1):D325–D334. doi:10.1093/nar/gkaa1113

- Köhler S, Carmody L, Vasilevsky N, Jacobsen JOB, Danis D, Gouridine JP, et al. Expansion of the Human Phenotype Ontology (HPO) knowledge base and resources. *Nucleic Acids Res.* 2019;47(D1):D1018–D1027. doi:10.1093/nar/gky1105

<!-- HPO database — used for tissue mapping and therapeutic window inference -->

Regulatory Guidance Documents

- US Food and Drug Administration. Long term follow-up after administration of human gene therapy products: guidance for industry. Silver Spring: FDA; 2020.
- US Food and Drug Administration. Chemistry, manufacturing, and control (CMC) information for human gene therapy investigational new drug applications (INDs): guidance for industry. Silver Spring: FDA; 2020.
- European Medicines Agency. Guideline on the quality, non-clinical and clinical aspects of gene therapy medicinal products. EMA/CAT/80183/2014. Amsterdam: EMA; 2018.

Disease and Dataset Source Audit

The 40-disease cohort is stored in `data/disease_cohort_40.csv`. Before final submission, each disease row must be checked against Orphanet and OMIM and then supported by at least one disease-specific review or primary paper where that disease is discussed in the Results/Discussion text. Do not cite a disease-specific claim unless the corresponding paper has actually been read and annotated.

Priority disease-source checks for the dissertation:

- Positive/benchmark controls: Hemophilia B, SMA, RPE65-associated LCA, ADA-SCID.
- Main high-scoring lysosomal/leukodystrophy cluster: MPS I, MPS II, MPS IIIA, metachromatic leukodystrophy, Krabbe disease, Gaucher disease, Fabry disease, Salla disease, mucopolidosis IV, alpha-mannosidosis.
- Liver metabolic cluster: MSUD, OTC deficiency, phenylketonuria, methylmalonic acidemia, glycogen storage disease Ia, Crigler-Najjar syndrome type I.
- Stress-test/scope-boundary diseases: DMD and LHON.

Minimum rule for the dissertation: every paragraph that interprets a disease cluster needs at least one checked source from this bibliography or from a disease-specific paper you have read.